$$P(y) = \int P(y|x)\, P(x)\, \mathrm{d}x$$

$$x_{n+1} = \lambda x_n (1 - x_n)$$

$$\lim_{x \to \infty} \frac{1}{1 + x^{-1}}$$

$$\sum_{n=1}^{N} n(n-1)$$

$$\int_b^a f(x)\, \mathrm{d}x$$

Introduction to LaTeX

Session 1: Introduction

Outline

1. What is LaTeX and why use it?
2. How does it work?
3. Document structure
4. Basic document formatting
5. Text formatting
6. Commenting
7. Diagnosing compilation errors

1

What is LaTeX and why use it?

What is LaTeX?

- TEX is a typesetting system. Originating in the 1980s, it was designed with maths at its core.
- LaTeX is the extension of this, designed for organising documents as preprints, articles, letters, etc.
- LaTeX has become the standard format for scientific writing.

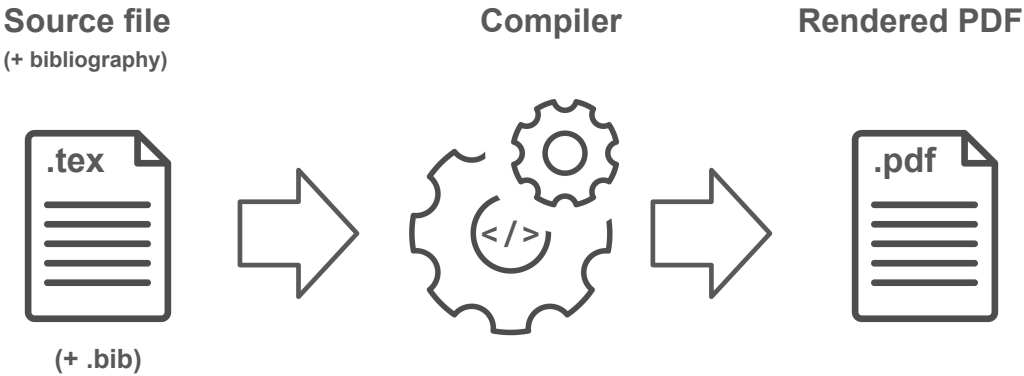
Why use LaTeX?

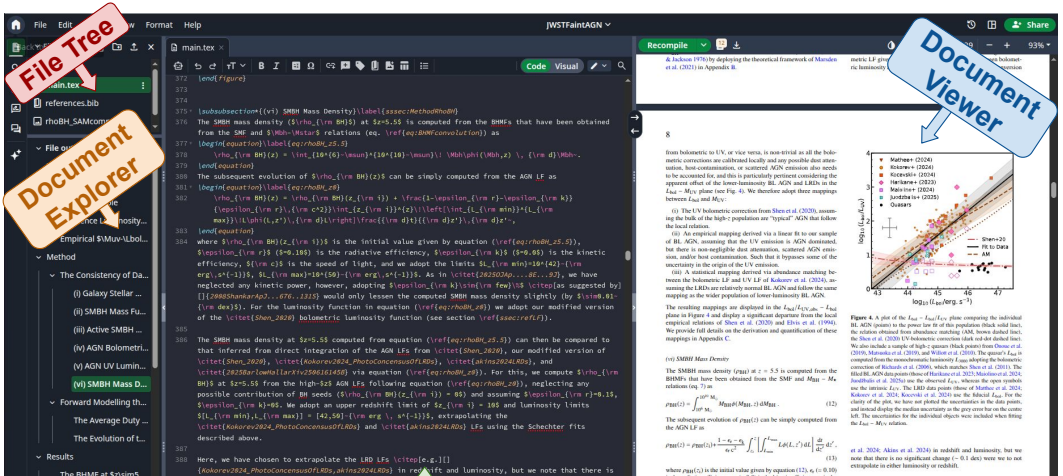
- **Maths:** Typeset equations clearly and correctly.
- **Formatting:** It makes formatting equations, figures, tables, citations easy
- **Reproducibility:** Files render identically, independent of OS
- **Referencing:** Organises and formats references and citations for you
- **Consistency:** Style and format are handled by `.sty` and `.cls`, so you're always consistent
- **Collaboration:** Online LaTeX platforms make collaborating and reviewing easy
- **Also,** some journals require LaTeX source files now

2

How does LaTeX work?

Compiling a document




File Tree
Document Explorer
Text Editor
Document Viewer
main.tex
references.bib
rhoBH_SAMcomp
File or
Method
The Consistency of Da...
(i) Galaxy Stellar ...
(ii) SMBH Mass Fu...
(iii) Active SMBH ...
(iv) AGN Bolometri...
(v) AGN UV Lumin...
(vi) SMBH Mass D...
Forward Modelling th...
The Average Duty ...
The Evolution of t...
Results
The BHMF at $z\sim5....
The Bolometric Lumin...
The UV Luminosity Fu...
The SMBH Mass Dens...
The SMBH Mass D...
The SMBH Mass D...
Code
Visual
JWSTFaintAGN
Recompile
Share
Matthee+ (2024)
Kokorev+ (2024)
Kocevski+ (2024)
Harikane+ (2023)
Inayoshi+ (2024)
Juodzbalis+ (2025)
Quasars
Shen+20
Fit to Data
AM
8

Overleaf[1] is the most common way to write in LaTeX as it:

- facilitates collaboration easily (sharing, reviewing, commenting)
- has a large library of packages and templates
- can integrate your reference manager (Mendeley, Zotoro, etc.)
- can submit directly to journals

You will already have access to the "premium" features via your uni account.

[1] https://www.overleaf.com/

https://www.overleaf.com/

A recommended local LaTeX installation is TeX Live (AKA MacTeX on macOS).

- Windows: Download install-tl-windows.exe and run it to install.
- macOS: Download MaxTex.pkg and run it to install.
- Linux:
  - For a light install, run
    `sudo apt install texlive-latex-extra texlive-fonts-recommended`
  - For a full install, run
    `sudo apt install texlive-full`

- To write the `.tex` file, you need a text editor (e.g. VS Code, Atom, Zed, gedit, etc.)
- You can compile the `.tex` to a `.pdf` in the command line by running

```
pdflatex [filename].tex
```

- VS Code has a useful extension called "LaTeX Workshop" which compiles whenever you save the `.tex`

3

Document Structure

```latex
% Document class
\documentclass{article}
% Preamble
\usepackage{graphicx}
% Document Information
\title{My Latex Document}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle
    % Main body
    Simple example document...

\end{document}
```

Document Class

Sets the core layout / style rules

Preamble

Gives further information and imports packages

Document Information

Metadata used in titles, headers, etc.

Main Body

The actual source code and text that appears in the document

\documentclass[options]{class}

`class` tells the compiler what kind of document to render; it's the name for a class (`.cls`) file (e.g. book, article, report, letter)

Most journals have their own `.cls` and/or `.sty` file that handles the formatting.

`options` tells the compiler about specific formatting options (e.g. paper size, font size, etc.)

Example:

```
\documentclass[11pt,oneside,onecolumn]{article}
```

Preamble

The preamble sits between the `\documentclass` line and the `\begin{document}` line to:

- Give further information about preferences

  e.g. format section headers
  `\titleformat{\section}[block]{\bfseries\color{Plum}}{\thetitle}{.5em}{#1}`

- Import packages

  e.g. including graphics `\{graphicx}`
  or formatting links `\usepackage{hyperref}`

- Define macros / commands

  e.g defining common variables or units
  `\newcommand \norm[1][2]{\mathcal{N}(#1,#2)}`

Packages

Much of LaTeXfunctionality is available in third-party packages.
Packages are included in the preamble with `\usepackage{name}`
Some common packages are:

- `amsmath`: Useful maths package from the Americam Maths Society
- `graphicx`: Handles the inclusion of figures and images
- `hyperref`: Handles the inclusion and formatting of links
- `natbib`: Bibliography manager (or, alternatively, biblatex)
- `geometry`: Handles page formatting

The \title{}, \author{}, \affil{}, \date{} commands provide metadata for the title, author list, headers, etc.

\title[ShortTitle]{FullTitle} – ShorTitle is used in the header

\author{AuthorName}

\affil[Number]{InstitutionName} – via the authblk package

\date{\today} – \today will auto-update the date for you

All the text and source code between `\begin{document}` and `\end{document}` lines is the main body of the document.

```
\begin{document}
\maketitle

    \begin{abstract}
        [Abstract]
    \end{abstract}

    \section{Introduction}
        [Text]

    \section{Method}
        [Text]

    [...]

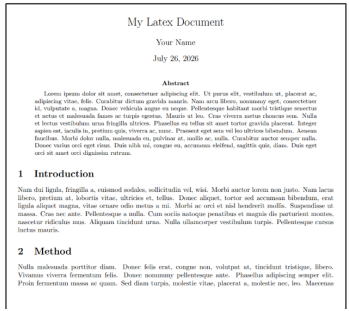
    \bibliography{References.bib}
\end{document}
```

Prints the title, author(s), date following the style sheet

Include the abstract

Split the text into sections

Include the bibliography


My Latex Document

Your Name

July 26, 2026

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1 Introduction

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

2 Method

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas

4

Basic Document Formatting

`\maketitle` – generates the title at the beginning of the document

`\tableofcontents` – generates a table of contents

`\section` – creates a new section

`\bibliography` – generates the bibliography

`\appendix` – denotes the start of the appendices

```
\part{name}
\chapter{name}
\section{name}
\subsection{name}
\subsubsection{name}
\paragraph{name}
\subparagraph{name}
```

For most projects `\section{}` is the top-level document command

All can be made unnumbered, so they don't appear in the TOC via (e.g.) `\section*{}`

5

Basic Text Formatting

LaTeXhas simple commands for common formatting options

Bold text via `\textbf{}` or `{\bf }`

Italic text via `\textit{}` or `{\it }`

Roman text via `\textrm{}` or `{\rm }`

`Typewriter` via `\texttt{}` or `{\tt }`

SmallCaps via `\textsc{}` or `{\sc}`

Underlined text via `\underline{}`

~~Strike through~~ via `\sout{}` from `ulem`

Subscripts via `\textsubscript{}` or `$_{...}$`

Superscripts via `\textsuperscript{}` or `$^{...}$`

```
\begin{itemize}
    \item One
    \item Two
    \item Three
\end{itemize}
```

⇓

- One
- Two
- Three

```
\begin{enumerate}
    \item One
    \item Two
    \item Three
\end{enumerate}
```

⇓

1. One
2. Two
3. Three

```
\begin{enumerate}
    \item One
 \begin{enumerate}
  \item One part a
  \item One part b
 \end{enumerate}
    \item Two
\end{enumerate}
```

⇓

1. One
   - A. One part a
   - B. One part b
2. Two

6 Commenting

Ways to comment

There are a few ways to add comments:

- In the soure code
- In the `.pdf`
- Via Overleafs review mode

Commenting in the source code

There are a few ways to add comments in the source code

- Single line comments

  ```
  % this is a comment
  ```

- There isn't built in multi-line comments, but one can comment-out a block of text by either

  Highlight the text and press `Ctrl + /` to add % to each line

  Or

  You can bodge it with

  ```
  \iffalse
    This is a multi-line comment
  \fi
  ```

It can be useful to define macros to add comments / notes to yours or others' work.

Adding a comment:

```
\def \dr#1{{\it\textcolor{teal}{DR:#1}}}
\dr{My comment...} ⇒ DR: My comment...
```

Adding a correction:

```
\def \corr #1#2{{\sout{\textcolor{red}{#1}}}{\,
{\textcolor{green}{#2}}}}
code\corr{wrong text}{correct text} ⇒ wrong text correct text
```

7

Common Errors When Compiling

Common errors

`Undefined control sequence` – Missing `\usepackage{}` declaration

`File '...' not found.` – Missing file or wrong path

`Missing \begin{document}` – Text in preamble prior to `\begin{document}`

`\begin{X} on input line ## ended by \{Y}` – Mismatched environment

`Missing $ inserted` – Maths / symbols outside math mode

`Too many}'s` – Missing { or extra}

`Runaway argument` – Expected a closing brace, but found a blank line.

`Undefined Citation` – Reference missing from `.bib`

(LaTeX StackExchange[2] is great for troubleshooting complex errors)

[2] https://tex.stackexchange.com/

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8

Hands-on Exercise

Log into Overleaf and create a simple document with:

- Title
- Author
- Sections and subsections
- Lists
- Formatted text

9

Homework

Expand your document to one page.



Extra Slides

You can add maths to text using inline math mode (more on this in session 3).

This can be done as:

... `$y=x^2$` ...
... `\(y=x^2\)` ...
... `\begin{math}y=x^2\end{math}` ...

All of which produce: ... $y = x^2$ ...

There are several citation styles:

`\citet{reference}` ⇒ Author (Year)

`\citep[pre-matter][post-matter]{reference}` ⇒ (pre-matter Author, Year, post-matter)

`\citealt{reference}` ⇒ Author Year

You can also cite multiple references in a single command, e.g.

`\citep[seealso][]{ref1,ref2}`⇒ (see also Author1, year1; Author2, year2)

More on this in session 3.

You can insert equations, figures and tables via (e.g.)

```
\begin{equation}
  f(x) = ax^2 + bx + c
\end{equation}
```

Overleaf will autofill the skeleton structure for figures and tables.
You will learn more details in sessions 2 and 3.

Many common symbols are included in LaTeX as standard (e.g. $\alpha, \beta, \gamma, \delta, \pm, \equiv, \sim, \simeq, \partial, \infty, \forall, \in, \sum, \wedge, ...$)

These can be inserted via common sense commands such as `\alpha`, `\beta`, ...

Extra symbols are available through packages such as asmsymb

Undefined control sequence

```
! Undefined control sequence
l.45 \includegraphics
      {figure.png}
```

Causes:

- Missing \package{} declaration
- Typo in the command

Common fixes:

- For \includegraphics: Add \usepackage{graphicx} to the preamble
- For maths: Add \usepackage{amsmath} to the preamble
- For colours:Add \usepackage{xcolor} to the preamble
- Check for typos in the command

File not found

```
! LaTeX Error: File ‘figures/example.png’ not found.
```

Causes:

- LaTeX cannot find the referenced file.
- This could be due to a wrong path in the argument, the file doesn't exist, or your compiling in the wrong directory (i.e. wrong relative path)

Common fixes:

- Check the path
- Check the file exists
- Check the PWD

Missing \begin{document}

```
! LaTeX Error: Missing \begin{document}
```

Causes:

- LaTeX found content before the start of the document because:
  - Typo in `\begin{document}`
  - There's content in the preamble

Common fixes:

- Check `\begin{document}` is after all `\usepackage{}` calls and other declarations
- Make sure no text appears before `\begin{document}`

Mismatched Environment

`! LaTeX Error: \begin{X} on line ## ended bu \end{Y}`

Causes:

- Environment X was opened by environment Y was closed
  - Typo in the environment name
  - Nested environments closed in the wrong order

Common fixes:

- Find the line and match every `\begin` with and `\end` working from in to out

Missing $ inserted

`! Missing $ inserted`

Causes:

- LaTeX has found maths (or what it thinks is maths) outside of math mode.
  - Maths without $...$ or $$...$$
  - unescaped underscore or caret
  - Greek letters or symbols outside math mode

Common fixes:

- Escape all underscores and carets (e.g. _ and $\hat{}$ to _ and \$\hat{}$)
- Wrap all maths, greek letters, symbols

Too many }'s

```
! Too many }'s
l.## }
```

Causes:

- Extra } somewhere
- Missing { prior to line ##

Common fixes:

- Go to line ##, count the braces and work back from there

```
Runaway argument?
! Paragraph was ended before \caption was complete
```

Causes:

- LaTeX expected a closing brace but found a blank line first, e.g.
  - Blank line in an argument
  - Missing closing brace

Common fixes:

- Go to the line, remove any blank lines, make sure all braces are closed

`LaTeX warning: Citation 'Einstein1905' on page # undefined`

Causes:

- Citing something that isn't in the bibliography file, e.g.
  - Typo in the reference name
  - Missing reference in `.bib`
  - Wrong bibliography file name

Common fixes:

- `Ctrl + f` to search for the reference and make sure that the key matches the `.bib` file
- Verify the correct `.bib` file is referenced in `\bibliography`
- Explicitly compile the `.bib` to `.bbl`